

Topic 2 [Types of Inheritance]:

Learn and Test different types of inheritance in C++. In each inheritance draw the class diagram with class chain and try to access the data members of bases classes from child classes.

i)Single Level Inheritance:

<pre>class A { private: int x; protected: int y; public: int z; }</pre>	<pre>class B:public A { //write public method to access // x,y & z }</pre>	<pre>int main() { B b; //call methods of class B return 0; }</pre>
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Code	
<pre>class A { private: int x; protected: int y; public: int z; }; </pre>	<pre>class B:public A { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} }; int main(){ B b; b.printx();b.printy();b.printz();} </pre>
Result:	<pre>C:\Users\MD... In member function 'void B::printx()': C:\Users\MD... 55 error: 'int A::x' is private within this context C:\Users\MD... 46 note: declared private here</pre>

ii)Multi-level Inheritance:

<pre>class A{ private: int x; protected: int y; public: int z; }</pre>	<pre>class B:public A{ }</pre>	<pre>class C:public B{ //write public //method to //access x,y & z }</pre>	<pre>int main(){ C c; //call //methods of //class C return 0 }</pre>
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Code	
<pre>class A { private: int x; protected: int y; public: int z; }; class B:public A { }; </pre>	<pre>class C:public B { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} }; int main() { C c; c.printx();c.printy();c.printz(); return 0; }</pre>
Result:	<pre>C:\Users\MD... In member function 'void C::printx()': C:\Users\MD... 58 error: 'int A::x' is private within this context C:\Users\MD... 46 note: declared private here</pre>

iii) Multiple Inheritance:

class A{ private: int x; protected: int y; public: int z; }	class B{ private: int p; protected: int q; public: int r; }	class C:public A, Public B{ //write public method //to access //x,y,z,p,q & r }	int main(){ C c; //call //methods of //class C return 0 }
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Code			
<pre>class A { private: int x; protected: int y; public: int z; }; class B { private: int p; protected: int q; public: int r; };</pre>		<pre>class C:public A,public B { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} void printp(){cout<<p<<" ";} void printq(){cout<<q<<" ";} void printr(){cout<<r<<" ";} }; int main() { C c; c.printx();c.printy();c.printz(); c.printp();c.printq();c.printr(); return 0; }</pre>	
Result:	<pre>C:\Users\MD... In member function 'void C::printx()': C:\Users\MD... 64 error: 'int A::x' is private within this context C:\Users\MD... 46 note: declared private here C:\Users\MD... In member function 'void C::printp()': C:\Users\MD... 67 error: 'int B::p' is private within this context C:\Users\MD... 55 note: declared private here</pre>		

iv) Heirarchical inheritance

class A{ private: int x; protected: int y; public: int z; }	class B:public A { //write public method to access x,y & z }	class C:public A { //write method public to access x,y & z }	int main(){ B b; C c; //call //methods of //class B & C return 0 }
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Code	
<pre>class A { private: int x; protected: int y; public: int z; }; class B:public A { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} };</pre>	<pre>class C:public A { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} }; int main() { B b; C c; b.printx();b.printy();b.printz(); c.printx();c.printy();c.printz(); return 0; }</pre>

Result:	C:\Users\MD...	In member function 'void B::printx()':
	C:\Users\MD... 55	error: 'int A::x' is private within this context
	C:\Users\MD... 46	note: declared private here
	C:\Users\MD...	In member function 'void C::printx()':
	C:\Users\MD... 62	error: 'int A::x' is private within this context
	C:\Users\MD... 46	note: declared private here

v) Hybrid (Diamond) inheritance [virtual class]

<pre> class A { private: int x; protected: int y; public: int z; } </pre>	<pre> class B:public A { } </pre>	<pre> class C:public A { } </pre>	<pre> class D:public B, public C { //write public method to access x,y & z } </pre>	<pre> int main(){ D d; //call //methods of //class D return 0 } </pre>
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Code											
<pre>class A { private: int x; protected: int y; public: int z; }; class B:public virtual A { }; class C:public virtual A { };</pre>	<pre>class D:public B, public C { public: void printx(){cout<<x<<" ";} void printy(){cout<<y<<" ";} void printz(){cout<<z<<" ";} }; int main() { D d; d.printx();d.printy();d.printz(); return 0; }</pre>										
Result:	<table><tr><td>C:\Users\MD...</td><td></td><td>In member function 'void D::printx()':</td></tr><tr><td>C:\Users\MD...</td><td>61</td><td>error: 'int A::x' is private within this context</td></tr><tr><td>C:\Users\MD...</td><td>46</td><td>note: declared private here</td></tr></table>	C:\Users\MD...		In member function 'void D::printx()':	C:\Users\MD...	61	error: 'int A::x' is private within this context	C:\Users\MD...	46	note: declared private here	
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